

Cases and Variables Practice

MATH 213

Group 1

- First, read the description of the study.
- Then answer the questions on the next slide.

Gamification and statistics, study components. Gamification is the application of game-design elements and game principles in non-game contexts. In educational settings, gamification is often implemented as educational activities to solve problems by using characteristics of game elements. Researchers investigating the effects of gamification on learning statistics conducted a study where they split college students in a statistics class into four groups: (1) no reading exercises and no gamification, (2) reading exercises but no gamification, (3) gamification but no reading exercises, and (4) gamification and reading exercises. Students in all groups also attended lectures. Students in the class were from two majors: Electrical and Computer Engineering ($n = 279$) and Business Administration ($n = 86$). After their assigned learning experience, each student took a final evaluation comprised of 30 multiple choice question and their score was measured as the number of questions they answered correctly. The researchers considered students' gender, level of studies (first through fourth year) and academic major. Other variables considered were expertise in the English language and use of personal computers and games, both of which were measured on a scale of 1 (beginner) to 5 (proficient). The study found that gamification had a positive effect on student learning compared to traditional teaching methods involving lectures and reading exercises. They also found that the effect was larger for females and Engineering students. ([Legaki et al. 2020](#))

Questions

1. Identify the main research question in this study.

EFFECTS OF GAMIFICATION ON LEARNING

1. Identify the subjects (cases) in this study. How many were included?

Cases: students

How many: 365

3. What variables were recorded in this study? Identify each as numerical or categorical.

Major (categorical), year (categorical), gender (categorical), gamification group (categorical), score (numerical), English expertise (categorical/numerical), computer expertise (categorical/numerical)

4. In the dataframe for this study, how many rows would there be? How many columns?

366 rows, 8 columns

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

Student	Major	Year	Gender	Group	English expertise	Computer expertise	Score
Jonathan	Business	3	M	2	5	4	24/30
Bronya	Computer	2	F	4	4	5	29/30

Group 2

- First, read the description of the study.
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Questions

1. Identify the main research question in this study.

How does the use of gamified educational resources affect student learning

1. Identify the subjects (cases) in this study. How many were included?

Cases: The Students

How many: 365

3. What variables were recorded in this study? Identify each as numerical or categorical.

Major: Categorical , Gender: Categorical, Test Score: Quantitative, Current academic year: Categorical, Test group: Categorical, English proficiency: Quantitative, Gaming proficiency: Quantitative

4. In the dataframe for this study, how many rows would there be? How many columns?

365 rows, 8 columns

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

Student	Major	Gender	Test Score	Year	Test group	English proficiency	Gaming proficiency
1	2	1	30	1	4	5	5
365	1	0	25	3	1	3	4

Group 3

- First, read the description of the study.
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Questions

1. Identify the main research question in this study.

Whether Gamification helps students learn better or if affects the students learning

1. Identify the subjects (cases) in this study. How many were included?

Cases: College Students

How many: 365

3. What variables were recorded in this study? Identify each as numerical or categorical.

Gamification and reading, the students took a 30 question multiple choice final evaluation

4. In the dataframe for this study, how many rows would there be? How many columns?

365 rows, 10 columns

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

[illegible]

Group 4

- First, read the description of the study.
- Then answer the questions on the next slide.

Gamification and statistics, study components. Gamification is the application of game-design elements and game principles in non-game contexts. In educational settings, gamification is often implemented as educational activities to solve problems by using characteristics of game elements. Researchers investigating the effects of gamification on learning statistics conducted a study where they split college students in a statistics class into four groups: (1) no reading exercises and no gamification, (2) reading exercises but no gamification, (3) gamification but no reading exercises, and (4) gamification and reading exercises. Students in all groups also attended lectures. Students in the class were from two majors: Electrical and Computer Engineering ($n = 279$) and Business Administration ($n = 86$). After their assigned learning experience, each student took a final evaluation comprised of 30 multiple choice question and their score was measured as the number of questions they answered correctly. The researchers considered students' gender, level of studies (first through fourth year) and academic major. Other variables considered were expertise in the English language and use of personal computers and games, both of which were measured on a scale of 1 (beginner) to 5 (proficient). The study found that gamification had a positive effect on student learning compared to traditional teaching methods involving lectures and reading exercises. They also found that the effect was larger for females and Engineering students. ([Legaki et al. 2020](#))

Questions

1. Identify the main research question in this study.
 - The study was conducted to find how students learning was impacted with gamification compared to traditional learning methods.
2. Identify the subjects (cases) in this study. How many were included?

Cases:

- Computer Engineering cases: 279 students
- Business Administration cases: 86 students

How many: 365 total cases

3. What variables were recorded in this study? Identify each as numerical or categorical.

- The Gender of the students (categorical)
- Academic Major (categorical)
- Year of student (categorical)
- Group number (categorical)
- English proficiency (categorical)
- Use of personal computers and games (categorical)
- Test Results (numerical)

4. In the dataframe for this study, how many rows would there be? How many columns?

- 365 rows (1 per student) x 8 columns

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

Student number	Gender	Major	Year	Group number	English proficiency	Use of personal computers and games	Test Results
Student 1	M	BA	1	1	3	4	28
Student 2	F	EE	2	2	4	5	20

Group 5

- First, read the description of the study.
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Gamification and statistics, study components. Gamification is the application of game-design elements and game principles in non-game contexts. In educational settings, gamification is often implemented as educational activities to solve problems by using characteristics of game elements. Researchers investigating the effects of gamification on learning statistics conducted a study where they split college students in a statistics class into four groups: (1) no reading exercises and no gamification, (2) reading exercises but no gamification, (3) gamification but no reading exercises, and (4) gamification and reading exercises. Students in all groups also attended lectures. Students in the class were from two majors: Electrical and Computer Engineering ($n = 279$) and Business Administration ($n = 86$). After their assigned learning experience, each student took a final evaluation comprised of 30 multiple choice question and their score was measured as the number of questions they answered correctly. The researchers considered students' gender, level of studies (first through fourth year) and academic major. Other variables considered were expertise in the English language and use of personal computers and games, both of which were measured on a scale of 1 (beginner) to 5 (proficient). The study found that gamification had a positive effect on student learning compared to traditional teaching methods involving lectures and reading exercises. They also found that the effect was larger for females and Engineering students. ([Legaki et al. 2020](#))

Questions

1. Identify the main research question in this study.

What effect does gamification have on learning

1. Identify the subjects (cases) in this study. How many were included?

Cases: Electrical and computer engineering students, and business students

How many: 279 Electrical and computer, 86 business students

3. What variables were recorded in this study? Identify each as numerical or categorical.

Reading, Gamification, Computer Usage, Gender, Year, English Proficiency, Major, Games.

Numerical: Student #, Score

Categorical: Gamification, Computer Usage, Gender, Year, English Proficiency, Major, games

4. In the dataframe for this study, how many rows would there be? How many columns?

365 Rows 7 Columns

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

	Major	Gamification level	Computer usage	Gender	Year	English Level	Games	Score
Student 1								
Student 2...								

Group 6

- First, read the description of the study.
- Then answer the questions on the next slide.

Gamification and statistics, study components. Gamification is the application of game-design elements and game principles in non-game contexts. In educational settings, gamification is often implemented as educational activities to solve problems by using characteristics of game elements. Researchers investigating the effects of gamification on learning statistics conducted a study where they split college students in a statistics class into four groups: (1) no reading exercises and no gamification, (2) reading exercises but no gamification, (3) gamification but no reading exercises, and (4) gamification and reading exercises. Students in all groups also attended lectures. Students in the class were from two majors: Electrical and Computer Engineering ($n = 279$) and Business Administration ($n = 86$). After their assigned learning experience, each student took a final evaluation comprised of 30 multiple choice question and their score was measured as the number of questions they answered correctly. The researchers considered students' gender, level of studies (first through fourth year) and academic major. Other variables considered were expertise in the English language and use of personal computers and games, both of which were measured on a scale of 1 (beginner) to 5 (proficient). The study found that gamification had a positive effect on student learning compared to traditional teaching methods involving lectures and reading exercises. They also found that the effect was larger for females and Engineering students. ([Legaki et al. 2020](#))

Questions

1. Identify the main research question in this study.

The main question was how gamification affected learning statistics in college students

1. Identify the subjects (cases) in this study. How many were included?

Cases: College students in the statistics class

How many: 365

3. What variables were recorded in this study? Identify each as numerical or categorical.

There were 7 variables recorded. Gamification group, academic major, gender, level of study, expertise in english, and proficiency with computers/gaming would all be categorical variables. The numerical variable we are measuring is the Score the student receives on the test at the end of the study.

4. In the dataframe for this study, how many rows would there be? How many columns?

There would be 365 rows and 7 columns in the dataframe for this study

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

Case	Gamification Group	Major	Gender	Level of Study	Expertise in English Language	Proficiency w/ Computers	Score
Std1	1	Comp sci	Male	3rd year	3	4	24/30
Std2	3	BM	Femal	2nd	4	3	26/30

Group 7

- First, read the description of the study.
- Then answer the questions on the next slide.

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Questions

1. Identify the main research question in this study.

What are the effects of gamification in college classes compared to traditional methods?

2. Identify the subjects (cases) in this study. How many were included?

Cases:

Engineering students (computer and electrical) and business administration students

How many:

279 (engineering students) 86 (business admin), 365 total

3. What variables were recorded in this study? Identify each as numerical or categorical.

Numerical: # of questions answered correctly (out of 30)

Categorical: Study/Experimental groups, gender, level of study, college major, level of familiarity in English, use of personal computers/games

4. In the dataframe for this study, how many rows would there be? How many columns?

365 rows, 7 columns (not including student name/identifier)

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

X	# of questions answered correctly	Study group	Gender	Level of Study	Major	English Proficiency	PC/ Game Proficiency
1	25	1	F	2	Electrical Engineering	4	5
2	17	4	M	3	Business Admin	5	4

Group 8

- First, read the description of the study.
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Questions

1. Identify the main research question in this study.

How does gamification compare to traditional teaching methods in a statistics class?

1. Identify the subjects (cases) in this study. How many were included?

Cases:

The entire statistics class. (300+ students)

How many:

4 cases

3. What variables were recorded in this study? Identify each as numerical or categorical.

Gender, academic level(1-4), academic major(EE,CE,Business), proficiency in the English language, and level of gamification and reading exercises

Categorical, numerical, categorical, numerical, categorical

4. In the dataframe for this study, how many rows would there be? How many columns?

300+ rows, 5 cols

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

	Gender	Academic level	Academic major	Proficiency in english	Test results	Game	
S1	M	2	CE	95	?		
S2	F	3	EE	99	?		

Group 9

- First, read the description of the study.
- Then answer the questions on the next slide.

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Questions

1. Identify the main research question in this study.

1. Identify the subjects (cases) in this study. How many were included?

Cases:

How many:

3. What variables were recorded in this study? Identify each as numerical or categorical.

4. In the dataframe for this study, how many rows would there be? How many columns?

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!

Group 10

- First, read the description of the study.
- Then answer the questions on the next slide.

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Questions

1. Identify the main research question in this study.

1. Identify the subjects (cases) in this study. How many were included?

Cases:

How many:

3. What variables were recorded in this study? Identify each as numerical or categorical.

4. In the dataframe for this study, how many rows would there be? How many columns?

5. In the table below, fill in the variable names in the appropriate spots. Then, make up some hypothetical data for two hypothetical cases, entering it into the table. Make sure you are using “tidy data” conventions!
